INCH-POUND

MIL-P-70715 (AR)
AMENDMENT 2
8 February 1991
SUPERSEDING
AMENDMENT 1
20 July 1988

MILITARY SPECIFICATION

PROJECTILE, 60MM, ILLUMINATING, XM721
METAL PARTS FOR

This amendment forms a part of Military Specification MIL-P-70715 (AR), dated 28 August 1987, and is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Title, Add "(M721)" after "XM721"; and wherever XM721 appears in the specification.

PAGE 2

2.1.3, add as follows:

"AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B46.1 - Surface Texture

(Application for copies should be addressed to: American National Standards Institute (ANSI), 1430 Broadway, New York, NY 10018)"

PAGE 5

Table I, change "934530" to read "9345350"

PAGE 6

Table I, change "Disc, Ignition" to read "Canister, Illuminant"
Add to Table I the following:

"Parachute Folding and Packing Assembly (dwg. 9345345) Examination for defects ... 10 ... 3.2 ... 4.4.2.15

Bag, Deployment and Chute, Drogue (dwg. 9345347) Examination for defects ... 10 ...3.2 ... 4.4.2.16

AMSC N/A FSC 1310

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Spring, Parachute Expulsion (Dwg. 9345343) Examination for defects ... 10 ... 3.2 ...4.4.2.17

Canister, Illuminant Assembly (Dwg. 9390652) Examination for defects ... 10 ... 3.2 ...4.4.2.18"

PAGE 8

4.4.2.1, Major 104, change "slot" to "groove"

4.4.2.1, Major 105, change "slot" to "groove"

4.4.2.1, Major 108, change to read "Distance from end of flange to forward face."

PAGE 10

4.4.2.2, Major 112, change to read "Length from forward face to rear outside shoulder."

4.4.2.2, Major 118, change to read "Total runout of rear OD with axis of ID"

4.4.2.2, Major 119, change inspection method from "Gage" to "Visual/4.5.4"

4.4.2.2, Major 120, change to read "Inside diameter of counterbore"

4.4.2.2, Major 122, change inspection method from "Gage" to "Test/4.5.2"

4.4.2.2, Minor 203, delete in its entirety and substitute, "Surface finish improper ... 0.65% ... 3.2 ... Visual/4.5.4"

4.4.2.2, add Minor 204 which reads as follows:

"Distance from forward face to obturator groove ... 0.65% ... 3.2 ... Gage"

4.4.2.2, add Minor 205 which reads as follows:

"Distance from rear face to rear shoulder ... 0.65% ... 3.2 ... Gage"

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4.4.2.2, add Minor 206 which reads as follows:

"Distance from forward face to bottom of counterbore ... 0.65% ... 3.2 ... Gage"

PAGE 10

4.4.2.2, add Minor 207 which reads as follows:

"Diameter of thread relief, min ... 3.2 ... Gage"

4.4.2.2, add Minor 208 which reads as follows:

"Evidence of poor workmanship ... 0.65% ... 3.4 ... Visual"

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4.4.2.3, Minor 201, delete in its entirety and substitute the following:

"Surface finish improper ... 0.65% ... 3.2 ... Visual/4.5.4"

4.4.2.3, add Minor 202 which reads as follows:

"Evidence of poor workmanship ... 0.65% ... 3.4 ... Visual"

PAGE 12

4.4.2.4, Minor 201, delete in its entirety and substitute the following:

"Radius at ID of well and bottom of well improper ... 0.65% ... 3.2 ... Gage"

4.4.2.4, add Minor 202 which reads as follows:

"Evidence of poor workmanship ... 0.65% ... 3.4 ... Visual"

4.4.2.5, delete Majors 101 through 109 in their entirety and substitute the following:

"101 Outside diameter above base ... 0.40% ... 3.2 ... Gage
102 Height of base ... 0.40% ... 3.2 ... Gage"

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4.4.2.5, delete Minors 201 through 203 in their entirety and substitute the following:

"201 Outside diameter of base of pin ... 0.65% ... 3.2 ... Gage
202 Location of hole on center of pin ... 0.65% ... 3.2 ... Gage
203 Distance from top to center of hole, min. ... 0.65% ... 3.2 ...Gage
204 Diameter of thru-hole, min. ... 0.65% ... 3.2 ... Gage
205 Overall height ... 0.65% ... 3.2 ... Gage
206 Surface finish improper ... 0.65% ... 3.2 ... Visual/4.5.4
207 Inside radius, max. ... 0.65% ... Gage
208 Evidence of poor workmanship ... 0.65% ... 3.4 ... Visual"

PAGE 14

4.4.2.6, Minor 201, change "base" to "flange"

PAGE 17

4.4.2.9, delete Minor 202 in its entirety.

PAGE 19

4.4.2.11, delete Minor 201 in its entirety and substitute the following:

"201 Radial cuts missing ... 0.65% ... 3.2 ... Visual
202 Evidence of poor workmanship ... 0.65% ... 3.4 ... Visual

PAGE 20

4.4.2.12, change Major 102 to read, "Inside diameter, max."

4.4.2.12, change Major 103 to read, "Wall thickness, min."

4.4.2.12, change Major 104 to read, "Diameter of thru-hole, max."

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4.4.2.13, delete Minor 201 in its entirety and substitute the following:

"201 Surface finish improper ... 0.65% ... 3.2 ... Visual/4.4.5
202 Evidence of poor workmanship ... 0.65% ... 3.4 ... Visual"

PAGE 22

4.4.2.14, change Major 101 to read, "Hole or tear in canopy greater than 1/8" diameter"

4.4.2.14, delete Major 104 in its entirety and substitute the following:

"Loop missing or improperly taped ... 0.40% ... 3.2 ... Visual"

4.4.2.14, change Major 107 to read, "Distance from edge of parachute to end of loop"

4.4.2.14, delete Minor 202 in its entirety.
Add the following new paragraphs (included):

"4.4.2.15 Parachute Folding and Packing Assembly (Dwg. 9345345)

4.4.2.16 Bag, Deployment and Chute, Drogue (Dwg. 9345347)

4.4.2.17 Spring, Parachute Expulsion (Dwg. 9345343)

4.4.2.18 Canister, Illuminant Assembly (Dwg. 9390652)"

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.15	Parachute Folding and Packing Assembly				9345345
					NEXT HIGHER ASSEMBLY
					9345340
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Parachute bag stitching missing or broken		0.40%	3.2	Visual
102	Drogue chute missing or damaged		0.40%	3.2	Visual
103	Improper accordian fold		0.40%	3.2	Visual
104	Shroud lines twisted or tangled or 1/2 of shroud lines not coiled under the last fold		0.40%	3.2	Visual
Minor					
201	Bag tie cord missing or improperly tied		0.65%	3.2	Visual
202	Shroud line projection out of bag		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.16	Bag Deployment and Chute, Drogue	1 OF 1	9345347
			NEXT HIGHER ASSEMBLY 9345345

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Drogue chute improperly attached to bag		0.40%	3.2	Visual/Manual
102	Drogue chute shroud line missing		0.40%	3.2	Visual
Minor					
201	Inside diameter of bag at open end		0.65%	3.2	Gage
202	Inside diameter of bag at bottom		0.65%	3.2	Gage
203	Inside depth of bag		0.65%	3.2	Gage
204	Stitching incorrect or broken		0.65%	3.2	Visual
205	Length and width of drogue chute canopy		0.65%	3.2	Gage
206	Shroud lines improperly stitched to drogue canopy		0.65%	3.2	Visual
207	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
4.4.2.17	Spring, Parachute Expulsion		1 of 1	9345343
				NEXT HIGHER ASSEMBLY: 9345340

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter of spring		0.40%	3.2	Gage
102	Free length		0.40%	3.2	Gage
Minor					
201	Length at solid height		0.65%	3.2	Gage
202	Ends squared and ground		0.65%	3.2	Visual
203	Wire diameter		0.65%	3.2	Gage
204	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.18	Canister, Illuminant Assembly	1	1	9390652

NEXT HIGHER ASSEMBLY: 12630532

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Canister illuminant assembly fails to pass through the ring gage		0.40%	3.2	Gage
102	Spot weld improper or missing		0.40% Note 1	3.2	Visual/Manual
Minor					
201	Overall length		0.65%	3.2	Gage
202	Protective finish missing or inadequate		0.65%	3.2	Visual
203	Poor Workmanship		0.65%	3.4	Visual

NOTES:

1. In addition to the visual inspection, five (5) parts from each lot shall be tested for adequate weld strength in accordance with paragraph 4.5.3.

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Add new paragraphs 4.5.3 and 4.5.4 as follows:

"4.5.3 Test for adequate weld strength. Test for adequate weld strength shall be conducted by pulling the support plate from the canister. No more than one undersize weld button is permitted on each sample tested. More than one weld button which measures less than 0.125 inches diameter shall be cause for rejection of the lot.

4.5.4 Surface roughness. The roughness comparison specimens prescribed by ANSI B46.1 shall be used as a basis for surface roughness determination. Four (4) power magnification may be used in performing the visual inspection."

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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